

CHIEDOZIE ONYEKWUM

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SUMMARY

Dedicated DevOps/MLOps engineer with expertise in cloud infrastructure, cybersecurity, and AI. Proven track record deploying secure, scalable systems across AWS and Azure. Pursuing an M.Sc. in Computer Science at Texas A&M, researching federated learning, machine unlearning, and autonomous systems. Focused on applying the right AI solution with disciplined attention to privacy, observability, and responsible deployment at scale.

EDUCATION

Texas A&M University - M.Sc., Computer Science

GPA: 3.85 | Expected Dec 2026

University of Port-Harcourt - B.Eng., Mechanical Engineering

TECHNICAL SKILLS

Languages: Go, Python, Bash, PowerShell, JavaScript, TypeScript

Containers & Virtualization: Docker, Kubernetes, Firecracker

IaC & CI/CD: Terraform, Pulumi, Terragrunt, Spacelift, Azure DevOps, GitHub Actions, CircleCI, Helm

Monitoring & ITSM: Prometheus, ELK Stack, Datadog, Fluent-bit, Jira, ServiceNow, PagerDuty

AI/ML & Web: Kubeflow, MLflow, Azure Databricks, LangChain, YOLOv8, LaneNet, Nginx

WORK EXPERIENCE

Samsung Austin Semiconductor

Austin, Texas, USA

AI/ML Engineer Co-op

Jan 2026 – Aug 2026

- Built an AI-driven opt-out detection feature that interprets free-form employee replies to incident alerts and automatically updates ban records, paired with alert-suppression and rate-limiting logic that reduced non-actionable incident alerts.
- Designed and built a full-stack ban/alerting portal (React, TypeScript, FastAPI, PostgreSQL) with SAML/SSO for employee onsite workforce; built a custom PowerShell CI/CD pipeline to deploy it in a secured, air-gapped environment, replacing a fully manual process with self-service access and full audit logging.

Texas A&M University

Texas, USA

Graduate Research Assistant

Apr 2025 – Dec 2025

- Conducted research and implemented solutions in machine unlearning, focusing on security, privacy, and broader AI applications.
- Researched PKI and Blockchain solutions to decentralize Certificate Authorities, improving security and high availability.
- Developed a proof-of-concept autonomous vehicle integrating RGB-LiDAR, Intel RealSense depth camera, GPS, and IMU on NVIDIA Jetson (Orin/Xavier); deployed YOLOv8 and LaneNet for real-time navigation.

Conclase

Colorado, USA

DevOps Engineer

Aug 2022 – Jan 2025

- Served as outsourced infrastructure engineer across healthcare and fintech client partnerships, owning end-to-end AWS infrastructure including ECS, EC2, CloudFormation, ALB, Route 53, S3, RDS, and CodePipeline/CodeBuild across multiple client environments.
- Architected a Feature Branch Testing (FBT) system using Terraform and AWS enabling developers to spin up fully isolated, disposable per-branch environments - significantly increasing testing frequency and reducing integration risk across client projects.
- Built a full environment replication pipeline (local - FBT - Dev - Staging - Prod) with full infrastructure parity, automated stage promotion, reusable Terraform modules, and per-environment secrets and config management.
- Hardened client infrastructure via CIS Benchmark adoption, automated certificate regeneration, AWS Firewall Manager consolidation, and rootless Docker runners; deployed secure SFTP server enabling encrypted B2B data exchange in regulated environments.
- Participated in 24/7 on-call rotation with defined SLA response windows, managing P1/P2 incidents via ServiceNow and PagerDuty across healthcare and fintech client infrastructure.
- Contributed to monolith-to-microservices migrations across client projects, reforming deployment strategies as application processes evolved - managing ECS service decomposition, ALB routing, blue/green and rolling deployments, and observability frameworks that maintained client SLOs and reduced disaster recovery time by 40% across full lifecycle stages.
- Automated Azure VM image pipeline that ingested on-premises build artifacts, triggered first-logon software provisioning scripts, and published versioned images to Azure Shared Image Gallery - giving internal teams ready-to-use environments instantly and eliminating manual copy overhead.

Gwrite

Port Harcourt, Nigeria

DevOps Engineer

Jan 2021 – Jun 2022

- Implemented HashiCorp Vault for Kubernetes secrets management and led SAP migration to Azure, achieving 70% RTO reduction.
- Architected secure containerization using Docker and CVSS tooling; transformed CI/CD processes with Tekton pipelines on Kubernetes.
- Pioneered AKS cluster setup following best practices and designed cost-optimised AWS infrastructure leveraging EC2, S3, RDS, and networking services.

Parkway Projects Africa

Lagos, Nigeria

Implementation / Software Engineer (Internship)

Sep 2019 – Oct 2021

- Developed serverless solutions on Azure/AWS with API and webhook integrations; managed cloud software incidents via ServiceNow.
- Built Django-based IT asset management application with real-time tracking; led network security implementation including VPN and ExpressRoute configurations.

PROJECTS & CONTRACTS

- **DozLab**: Cloud-native Kubernetes SaaS platform for DevOps/cybersecurity education; microservice architecture with sidecar containers, VM-based hands-on labs, and real-time WebSocket proxying. [[GitHub](#)]
- ECO-T (EDAT): Full-stack GHG accounting platform - see Leadership section for full detail.
- Sherlock: AI-powered Root Cause Analysis platform enhancing team collaboration and incident communication.
- ML Infrastructure: Optimised Kubernetes-based ML workflows using Kubeflow for pipeline orchestration and reproducibility.

CERTIFICATIONS

- **DevOps Engineer Expert** - Microsoft Azure [[20E994CB11398236](#)]
- **Azure Administrator** - Microsoft Azure [[9565CC0A4322528D](#)]
- **AWS Certified Cloud Practitioner** - Amazon Web Services (AWS) [[see credentials](#)]
- **Certified Incident Responder** - PagerDuty [[see credentials](#)]
- **Azure Fundamentals** - Microsoft Azure [[F0D7478D07C5C34C](#)]

LEADERSHIP

EDAT – VA (Independent Sustainability Assurance)

USA (Remote)

VP of Engineering (Volunteer)

Apr 2025 – Present

- Lead engineering team structure, recruitment, and technical direction for a sustainability assurance platform specializing in ESG verification, GHG validation, and voluntary carbon market auditing.
- Architected and built ECO-T - a full-stack GHG accounting platform covering Scope 1/2/3 emissions, API integrations with EPA, ERP systems, and energy grids, carbon pricing & offsets, and Sankey diagram production.
- Designed AI/ML pipelines for automated ESG verification and carbon disclosure across GRI, TCFD, CSRD, ISO 14064, VCS, and Gold Standard frameworks.
- Built backend APIs and DevOps/cloud infrastructure supporting GHG reporting automation, voluntary carbon market assurance, and third-party audit workflows.

National Society of Black Engineers, Texas A&M

Texas, USA

Vice President

Jan 2025 – Present

- Facilitate career development and cross-disciplinary collaboration through targeted projects that expand members' competitive edge and professional opportunities.

Microsoft

DC, USA

Student Ambassador

Jan 2018 – Nov 2021

- Mentored students in software development; developed school data analytics tools with Azure integration and organised community tech events.